HUJI Applicative Research Report — Exact & Social Sciences — September 2026

1 paper(s) · Exact & Social Sciences · Monthly · September 2026 · Generated September 01, 2026

■■ There is no high-potential applicable research this period — only lower-scoring papers were found. Presenting the two best below.

This month's overview highlights a significant advancement in materials science with direct implications for future optoelectronic technologies. Hebrew University researchers have achieved contact-free detection of a giant photovoltage in novel chiral 2D perovskites, demonstrating highly efficient light-to-electricity conversion. This breakthrough positions the technology for applications in ultra-sensitive optical sensors, secure communications, and advanced light-harvesting systems, addressing critical needs in diverse high-growth markets.

RESEARCHERS & RESEARCH SUBJECTS — BEST AVAILABLE

- **Igal Levine** — Materials, Clean Energy, Quantum

EXACT & SOCIAL SCIENCES — RESEARCH HIGHLIGHTS

1. Contactless Detection of Giant Helicity-Dependent Photovoltage in Chiral 2D Perovskites. 25/50

"Chiral Perovskites Unlock Contact-Free, High-Voltage Optical Sensing and Advanced Spintronics."

Main researcher: Igal Levine

Institute of Chemistry and the Center for Nanoscience and Nanotechnology, the Hebrew University of Jerusalem, Jerusalem, Israel.

Published in Small (Weinheim an der Bergstrasse, Germany) · 2026 Aug 8

ABOUT THIS RESEARCH

Researchers have developed a method to detect high-voltage electrical signals in chiral 2D perovskite materials using light with specific rotational polarity, all without physical contact. This discovery demonstrates a highly efficient way to convert circularly polarized light into electricity using asymmetric crystal structures.

COMMERCIAL ANGLE

Development of ultra-sensitive, non-contact optical sensors and next-generation spintronic devices for secure communications and advanced light-harvesting energy cells.

COMMERCIALISATION METRICS

Novelty 7/10

The integration of contactless detection with giant helicity-dependent photovoltage in 2D perovskites represents a significant technical leap over traditional contact-based measurements, though it builds on established chiral semiconductor physics.

Commercial Potential 4/10

While the discovery of giant photovoltage in chiral 2D perovskites has long-term potential for optoelectronics and sensing, the lack of a defined application or device architecture makes the immediate commercial path speculative.

Market Size 6/10

The discovery of giant photovoltage in chiral 2D perovskites suggests broad potential for next-generation optoelectronics, sensors, and quantum computing, though specific commercial applications remain early-stage.

Tech Readiness 2/10

The research focuses on fundamental physical phenomena (helicity-dependent photovoltage) in a novel material class, representing basic laboratory-stage discovery rather than an applied device or commercial prototype.

IP Strength 6/10

The innovation involves a specific material class (chiral 2D perovskites) and a novel detection method, which is patentable; however, defensibility is moderate as it relies on material composition and physical effects that may be bypassed by alternative chiral semiconductors.

View in Dashboard Open Paper